Thin Disk Hydrostatic Equilibrium Derivation

Assume

$$\varepsilon = \frac{H}{L} \ll 1. \tag{1}$$

For a thin disk,

$$V_z \ll V_r, V_\theta. \tag{2}$$

$$B_z \ll B_r, B_\theta. \tag{3}$$

Radial Momentum Balance

$$\rho_0 \left[\frac{\partial V_r}{\partial t} + (\mathbf{V} \cdot \nabla) V_r \right] = -\frac{\partial P_T}{\partial r} - \frac{GM\rho_0}{r^2} - \rho_0 \frac{\partial \phi_{\text{sg}}}{\partial r} + \frac{(\mathbf{B} \cdot \nabla) B_r}{\mu_0} + \rho_0 \Omega^2 r. \tag{4}$$

Vertical Momentum Balance

$$\rho_0 \left[\frac{\partial V_z}{\partial t} + (\mathbf{V} \cdot \nabla) V_z \right] = -\frac{\partial P}{\partial z} - \frac{GM\rho_0 z}{r^3} - \rho_0 \frac{\partial \phi_{\text{sg}}}{\partial z} + \frac{(\mathbf{B} \cdot \nabla) B_z}{\mu_0}. \tag{5}$$

Since $\varepsilon \ll 1$, the vertical acceleration terms are negligible, giving

$$\frac{\partial P}{\partial z} = -\frac{GM\rho_0\, z}{r^3}. \tag{6}$$

Vertical Pressure Profile

Integrating over z,

$$P = -\frac{GM\rho_0}{2r^3} z^2 + C. \tag{7}$$

Applying the free-surface boundary conditionat $z = H(r)$ $P = 0$,

$$C = \frac{GM\rho_0 H^2(r)}{2r^3}. \tag{8}$$

Hence

$$\boxed{P = \frac{GM\rho_0}{2r^3} \left[H^2(r) - z^2 \right].} \tag{9}$$

Depth-Averaged Pressure

The disk occupies $z \in [0, H]$. By symmetry of P in z, define

$$\bar{P} = \frac{1}{H} \int_0^H P \, dz. \tag{10}$$

Substituting the pressure profile,

$$\bar{P} = \frac{GM\rho_0}{2Hr^3} \int_0^H \left[H^2(r) - z^2 \right] dz. \tag{11}$$

Evaluating the integral,

$$\bar{P} = \frac{GM\rho_0}{2Hr^3}\left[H^2 z - \frac{z^3}{3}\right]_0^H \tag{12}$$

$$= \frac{GM\rho_0}{2Hr^3}\left[H^3 - \frac{H^3}{3}\right] \tag{13}$$

$$= \frac{GM\rho_0}{2Hr^3} \cdot \frac{2H^3}{3}. \tag{14}$$

Therefore,

$$\boxed{\bar{P} = \frac{GM\rho_0 H^2(r)}{3r^3}.} \tag{15}$$

Depth-Averaged Radial Momentum Equation

After vertical integration and using the thin-disk closure, the depth-averaged radial momentum equation is:

$$\rho_0\left[\frac{\partial(H\bar{V}_r)}{\partial t} + [(\mathbf{V}\cdot\nabla)V\,H]_r - 2\Omega V_\theta H\right] = -\frac{\partial(H\bar{P})}{\partial r} + \rho_0\Omega^2 rH - \frac{\rho_0 GMH}{r^2} - \rho_0\left(\frac{\partial\phi}{\partial r}\right)_{\mathrm{sg}} + \frac{[(\mathbf{B}\cdot\nabla)\mathbf{B}\,H]_r}{\mu_0}. \tag{16}$$

Note that planetary gravity does not appear explicitly here because it is already encoded in $\bar{P}$ through the hydrostatic vertical balance.

Base State

For the equilibrium state $V = 0$, $B = 0$. Neglecting self-gravity,

$$\left(\frac{\partial\phi}{\partial r}\right)_{\mathrm{sg}} = 0. \tag{17}$$

The radial momentum equation reduces to:

$$0 = -\frac{\partial P}{\partial r} + \rho_0\Omega^2 r - \frac{\rho_0 GM}{r^2}.. \tag{18}$$

Expanding the pressure gradient using the product rule,

$$\frac{\partial P}{\partial r} = \rho_0\Omega^2 r\; - \frac{\rho_0 GM}{r^2}. \tag{19}$$

Keplerian Rotation

For Keplerian rotation $\Omega^2 = GM/r^3$, the right-hand side vanishes:

$$\frac{2H}{r^3}\frac{dH}{dr} - \frac{3H^2}{r^4} = 0. \tag{20}$$

Dividing by $3H/r^3$:

$$\frac{dH}{dr} = \frac{H}{r} \cdot \frac{3}{2}. \tag{21}$$

$$H = H_0\left(\frac{r}{r_0}\right)^{\frac{3}{2}} \tag{22}$$

Derivation of the Linearised Matrix Equation

1. Initial Linearised Equations

Linearised equations about the background state $\vec{V}_0 = 0, \quad B_0 = 0, \quad H = H(r)$.

Continuity Equation:

$$\frac{\partial \eta}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(HrV_r') + \frac{1}{r}\frac{\partial}{\partial \theta}(HV_\theta') = 0 \tag{23}$$

r **Momentum (LMB) Equation:**

$$\frac{\partial}{\partial t}(HV_r') + GM\left[\frac{2H}{r^3}\frac{\partial H}{\partial r}\eta + \frac{H^2}{r^3}\frac{\partial \eta}{\partial r} - \frac{3H^2\eta}{r^4}\right] - 2\Omega HV_\theta' - \frac{B_0}{\mu_0\rho}B_r' + H\frac{\partial \phi'}{\partial r} = 0 \tag{24}$$

θ **Momentum (LMB) Equation:**

$$\frac{\partial}{\partial t}(HV_\theta') + \frac{GMH^2}{r^4}\frac{\partial \eta}{\partial \theta} + 2\Omega V_r' H - \frac{B_0}{\mu_0\rho}B_\theta' + \frac{H}{r}\frac{\partial \phi'}{\partial \theta} = 0 \tag{25}$$

r **Induction Equation:**

$$\frac{\partial}{\partial t}(HB_r') + B_0 V_r' = 0 \tag{26}$$

θ **Induction Equation:**

$$\frac{\partial}{\partial t}(HB_\theta') + B_0 V_\theta' = 0 \tag{27}$$

2. Substitution of the Scale Height Profile

Given $H = H_0\left(\frac{r}{r_0}\right)^{3/2}$, the radial derivative is:

$$\frac{\partial H}{\partial r} = \frac{3H}{2r} \tag{28}$$

Substituting this into the initial equations and dividing by H yields the simplified system:

Continuity:

$$\frac{\partial \eta}{\partial t} + H\frac{\partial V_r'}{\partial r} + \frac{5H}{2r}V_r' + \frac{H}{r}\frac{\partial V_\theta'}{\partial \theta} = 0 \tag{29}$$

r **Momentum:**

$$\frac{\partial V_r'}{\partial t} + GM\frac{H}{r^3}\frac{\partial \eta}{\partial r} - 2\Omega V_\theta' - \frac{B_0}{\mu_0\rho H}B_r' + \frac{\partial \phi'}{\partial r} = 0 \tag{30}$$

θ **Momentum:**

$$\frac{\partial V_\theta'}{\partial t} + \frac{GMH}{r^4}\frac{\partial \eta}{\partial \theta} + 2\Omega V_r' - \frac{B_0}{\mu_0\rho H}B_\theta' + \frac{1}{r}\frac{\partial \phi'}{\partial \theta} = 0 \tag{31}$$

Induction:

$$\frac{\partial B_r'}{\partial t} + \frac{B_0}{H}V_r' = 0 \tag{32}$$

$$\frac{\partial B_\theta'}{\partial t} + \frac{B_0}{H}V_\theta' = 0 \tag{33}$$

3. Application of the Ansatz

Applying the normal mode ansatz $q' = \tilde{q} \exp\left[i\left(\int k_r\, dr + m\theta - \omega t\right)\right]$ and the specified gravitational potential gradients:

$$\frac{\partial \phi'}{\partial r} \to -\frac{2\pi i G_N \rho\, k_r}{|k|}\tilde{\eta} \tag{34}$$

$$\frac{1}{r}\frac{\partial \phi'}{\partial \theta} \to -\frac{2\pi i G_N \rho}{|k|}\frac{m}{r}\tilde{\eta} \tag{35}$$

As requested, we neglect the explicit r-dependence introduced by the scale height derivative ($i\frac{5k_r}{2r} \to 0$) in the dispersion relation. The equations transform into the algebraic system:

$$-i\omega\tilde{\eta} + iHk_r\tilde{V}_r + \frac{imH}{r}\tilde{V}_\theta = 0 \tag{36}$$

$$-i\omega\tilde{V}_r + \frac{ik_r GMH}{r^3}\tilde{\eta} - 2\Omega\tilde{V}_\theta - \frac{B_0}{\mu_0\rho H}\tilde{B}_r - \frac{2\pi i G\rho k_r}{|k|}\tilde{\eta} = 0 \tag{37}$$

$$-i\omega\tilde{V}_\theta + \frac{imGMH}{r^4}\tilde{\eta} + 2\Omega\tilde{V}_r - \frac{B_0}{\mu_0\rho H}\tilde{B}_\theta - \frac{2\pi i G\rho m}{|k|r}\tilde{\eta} = 0 \tag{38}$$

$$-i\omega\tilde{B}_r + \frac{B_0}{H}\tilde{V}_r = 0 \tag{39}$$

$$-i\omega\tilde{B}_\theta + \frac{B_0}{H}\tilde{V}_\theta = 0 \tag{40}$$

4. Elimination of Magnetic Field Perturbations

Rearranging the induction equations provides the magnetic field variables in terms of velocity:

$$\tilde{B}_r = -i\frac{B_0}{\omega H}\tilde{V}_r, \quad \tilde{B}_\theta = -i\frac{B_0}{\omega H}\tilde{V}_\theta \tag{41}$$

Substituting these into the momentum equations and grouping terms for $\tilde{V}_r, \tilde{V}_\theta$, and $\tilde{\eta}$:

r **Momentum:**

$$ik_r\left(\frac{GMH}{r^3} - \frac{2\pi G_N\rho}{|k|}\right)\tilde{\eta} - i\left(\omega - \frac{B_0^2}{\mu_0\rho\omega H^2}\right)\tilde{V}_r - 2\Omega\tilde{V}_\theta = 0 \tag{42}$$

θ **Momentum:**

$$\frac{im}{r}\left(\frac{GMH}{r^3} - \frac{2\pi G_N\rho}{|k|}\right)\tilde{\eta} + 2\Omega\tilde{V}_r - i\left(\omega - \frac{B_0^2}{\mu_0\rho\omega H^2}\right)\tilde{V}_\theta = 0 \tag{43}$$

5. Matrix Formulation

Constructing the final system $M\mathbf{x} = 0$ for the primary variables $\mathbf{x} = [\tilde{\eta}, \tilde{V}_r, \tilde{V}_\theta]^T$ using the continuity and substituted momentum equations. To ensure proper fit on the page, the matrix is scaled.

$$\begin{bmatrix} -i\omega & iHk_r & \frac{imH}{r} \\ ik_r\left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|}\right) & -i\left(\omega - \frac{B_0^2}{\mu_0\rho\omega H^2}\right) & -2\Omega \\ \frac{im}{r}\left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|}\right) & 2\Omega & -i\left(\omega - \frac{B_0^2}{\mu_0\rho\omega H^2}\right) \end{bmatrix} \begin{bmatrix} \tilde{\eta} \\ \tilde{V}_r \\ \tilde{V}_\theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

6. Matrix Determinant Formulation

To find the non-trivial solutions, compute $\det(M) = 0$. For compactness, define the modified frequency $\tilde{\omega}$ and the gravitational potential modifier Φ:

$$\tilde{\omega} = \omega - \frac{B_0^2}{\mu_0 \rho \omega H^2} \tag{44}$$

$$\Phi = \frac{GMH}{r^3} - \frac{2\pi G \rho}{|k|} \tag{45}$$

Substituting these into the original matrix yields:

$$M = \begin{bmatrix} -i\omega & iHk_r & \frac{imH}{r} \\ ik_r\Phi & -i\tilde{\omega} & -2\Omega \\ \frac{im}{r}\Phi & 2\Omega & -i\tilde{\omega} \end{bmatrix} \tag{46}$$

The determinant expands as:

$$\det(M) = M_{11}(M_{22}M_{33} - M_{23}M_{32}) - M_{12}(M_{21}M_{33} - M_{23}M_{31}) + M_{13}(M_{21}M_{32} - M_{22}M_{31}) = 0 \tag{47}$$

7. Exact Dispersion Relation

Let $k^2 = k_r^2 + \frac{m^2}{r^2}$:

$$\omega \left[\left(\omega - \frac{B_0^2}{\mu_0 \rho \omega H^2} \right)^2 - 4\Omega^2 \right] - H \left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|} \right) \left[k^2 \left(\omega - \frac{B_0^2}{\mu_0 \rho \omega H^2} \right) + \frac{5m\Omega}{r^2} \right] = 0 \tag{48}$$

8. Polynomial Formulation

Define the Alfvén frequency squared:

$$\omega_A^2 = \frac{B_0^2}{\mu_0 \rho H^2} \tag{49}$$

The dispersion relation becomes:

$$\omega^3 - 2\omega_A^2 \omega + \frac{\omega_A^4}{\omega} - 4\Omega^2 \omega - H\Phi k^2 \omega + H\Phi k^2 \frac{\omega_A^2}{\omega} - H\Phi \frac{5m\Omega}{r^2} = 0 \tag{50}$$

Multiply the entire equation by ω to yield a quartic equation:

$$\omega^4 - \left[2\omega_A^2 + 4\Omega^2 + H\Phi k^2 \right] \omega^2 - \left(H\Phi \frac{5m\Omega}{r^2} \right) \omega + \omega_A^2 \left(\omega_A^2 + H\Phi k^2 \right) = 0 \tag{51}$$

The equation takes the standard polynomial form:

$$C_4 \omega^4 + C_3 \omega^3 + C_2 \omega^2 + C_1 \omega + C_0 = 0 \tag{52}$$

The expanded coefficients are:

$$C_4 = 1 \tag{53}$$

$$C_3 = 0 \tag{54}$$

$$C_2 = -\left[2\omega_A^2 + 4\Omega^2 + H\left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|}\right)k^2\right] \tag{55}$$

$$C_1 = -\frac{5m\Omega H}{r^2}\left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|}\right) \tag{56}$$

$$C_0 = \omega_A^2\left[\omega_A^2 + H\left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|}\right)k^2\right] \tag{57}$$

9. Deduction of Cases and the Rossby Mode

Non-Magnetic Case ($B_0 = 0$)

Setting $\omega_A = 0$, the C_0 term vanishes, and we can factor out ω:

$$\omega\left[\omega^3 - \left(4\Omega^2 + H\Phi k^2\right)\omega - H\Phi\frac{5m\Omega}{r^2}\right] = 0 \tag{58}$$

One root is trivially $\omega = 0$. The other roots satisfy the cubic:

$$\omega^3 - \left(4\Omega^2 + H\Phi k^2\right)\omega - H\Phi\frac{5m\Omega}{r^2} = 0 \tag{59}$$

If we consider the tightly wound limit ($k \gg m/r$), the high-frequency solutions are standard inertia-gravity (density) waves:

$$\omega^2 \approx 4\Omega^2 + H\Phi k^2 = 4\Omega^2 + H\left(\frac{GMH}{r^3} - \frac{2\pi G\rho}{|k|}\right)k^2 \tag{60}$$

The Rossby Mode Limit

The Rossby mode is characterized by low frequencies ($\omega \ll \Omega$). In the non-magnetic limit ($B_0 = 0$), we drop the ω^3 term in the cubic equation compared to the linear term:

$$-\left(4\Omega^2 + H\Phi k^2\right)\omega \approx H\Phi\frac{5m\Omega}{r^2} \tag{61}$$

Yielding the standard Rossby mode frequency:

$$\omega_{\text{Rossby}} \approx -\frac{H\Phi\frac{5m\Omega}{r^2}}{4\Omega^2 + H\Phi k^2} \tag{62}$$

This clearly shows the requirement for azimuthal dependence ($m \neq 0$) to support the Rossby wave. In the purely radial limit ($m = 0$), $\omega_{\text{Rossby}} \to 0$.

Magnetic Case ($B_0 \neq 0$)

With a non-zero magnetic field, $C_0 \neq 0$, and the true $\omega \to 0$ root is lifted. For low frequencies, we balance the lowest order terms in ω ($C_2\omega^2 + C_1\omega + C_0 \approx 0$):

$$-\left[2\omega_A^2 + 4\Omega^2 + H\Phi k^2\right]\omega^2 - \left(H\Phi\frac{5m\Omega}{r^2}\right)\omega + \omega_A^2\left(\omega_A^2 + H\Phi k^2\right) \approx 0 \tag{63}$$

Assuming $4\Omega^2$ is dominant in the quadratic coefficient (as $\Omega \gg \omega_A$), we have:

$$4\Omega^2\omega^2 + \left(H\Phi\frac{5m\Omega}{r^2}\right)\omega - \omega_A^2\left(\omega_A^2 + H\Phi k^2\right) \approx 0 \tag{64}$$

This quadratic equation provides two low-frequency roots, one of which represents a magnetically-modified Rossby mode, and the other is a new slow mode introduced by the magnetic tension. The presence of the $-C_0$ term can induce instabilities when $\omega_A^2 + H\Phi k^2 < 0$.